

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 21: Trigonometrical Identities

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Prove that $(1 - \sin^2 A) \sec^2 A = 1$.
2. Prove that $\tan^2 A - \sin^2 A = \tan^2 A \sin^2 A$.
3. Prove that $\frac{1}{1 + \sin A} + \frac{1}{1 - \sin A} = 2 \sec^2 A$.
4. Prove that $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$.
5. Prove that $(\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2$.
6. Prove that $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$.
7. Evaluate $\frac{\cos 38^\circ}{\sin 52^\circ} - \frac{\tan 19^\circ}{\cot 71^\circ}$.
8. Evaluate $\sin^2 34^\circ + \sin^2 56^\circ + 2 \tan 18^\circ \tan 72^\circ$.
9. If $\tan 2A = \cot(A - 18^\circ)$, where $2A$ is acute, find A .
10. If $A + B = 90^\circ$, prove that $\sqrt{\frac{\tan A \tan B + \tan A \cot B}{\sin A \sec B}} = \sec A$.

Answer Key

1. $\cos^2 A \times \frac{1}{\cos^2 A} = 1$

2. $\text{LHS} = \sin^2 A \left(\frac{1}{\cos^2 A} - 1 \right) = \sin^2 A \times \frac{\sin^2 A}{\cos^2 A}$

3. $\text{LHS} = \frac{2}{1 - \sin^2 A} = \frac{2}{\cos^2 A}$

4. Numerator = $\sin^2 A + 1 + 2 \cos A + \cos^2 A = 2(1 + \cos A)$; $\text{LHS} = \frac{2}{\sin A}$

5. $1 + 2 \sin A \cos A + 1 - 2 \sin A \cos A = 2$

6. $\text{LHS} = \sec^2 A (\sec^2 A - 1) = (1 + \tan^2 A) \tan^2 A$

7. $1 - 1 = 0$

8. $1 + 2 = 3$

9. $90^\circ - 2A = A - 18^\circ$, so $A = 36^\circ$

10. $\tan B = \cot A$, $\cot B = \tan A$, $\sec B = \text{cosec } A$:
inside the root = $\frac{1 + \tan^2 A}{1} = \sec^2 A$